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Semantic-Gravity Iterative Closest Point and AI-assisted navigation for Robot-Ready Construction Digital Twins ☆

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ABSTRACT

This paper introduces BIM2RDT (Building Information Models to Robot-Ready Construction Digital Twins), an AI-assisted pipeline transforming static BIM into dynamic digital twins for construction robot navigation and safety monitoring. The framework fuses BIM geometry, unmanned ground vehicle (UGV) visual observations, and IoT sensor data to enable automated site monitoring and safety-aware path planning. A core contribution is Semantic-Gravity ICP (SG-ICP), a point cloud registration method leveraging LLM-guided semantic reasoning to assign BIM-family-specific upright priors, improving alignment robustness in low-feature construction scenes. The framework integrates YOLOE open-vocabulary detection for BIM-conditioned perception and demonstrates safety monitoring through Hand-Arm Vibration (HAV) sensing written into IFC-compatible digital twin representations. Controlled experiments show SG-ICP reduces RMSE by approximately 60%–97% over standard registration methods in high-noise scenarios. These capabilities represent an initial step towards agentic construction workflows, with LLM-orchestrated pipelines that condition perception, trigger IFC updates, and propose navigation constraints.

1. Introduction

The Architecture, Engineering, and Construction (AEC) industry is in the midst of a profound digital transformation. Historically characterized as a late adopter of digital technology, the sector has traditionally relied heavily on the experiential knowledge of its professionals. However, growing pressures from economic uncertainties due to factors such as low productivity, persistent labor shortages, and the increasing complexity of modern projects are compelling companies to rapidly adopt digital construction management strategies to optimize resources and workflows. Market projections indicate that adoption of technologies such as the Digital Twins (DTs) in the construction market will reach $155.01 billion by 2030, with safety applications and worker monitoring driving significant growth [1]. For example, Cyber–Physical Systems (CPS) integrate computation, networking, and physical processes, creating a feedback loop where physical systems are monitored and controlled by computer-based algorithms [2]. Technological advances such as Building Information Modeling (BIM), DT, robotics, and Artificial Intelligence (AI) form the basis of a next-generation approach to construction management that moves beyond manual processes and reactive decision-making [3].

This paper presents BIM2RDT, **B**uilding **I**nformation **M**odels **to** **R**obot-**R**eady Construction **D**igital **T**wins, where the central '2' embodies both our BIM transformational approach and the dual 'R' requirements for robot readiness. BIM2RDT is a context-aware, AI-powered framework that transforms static BIM models into a digital twin, a medium that the robot can utilize in dynamic environments, with virtual components for robot path planning, activity recognition, and safety monitoring and compliance during construction execution. BIM2RDT uses the BIM as an *a priori* map to condition perception and constrain alignment, then writes geometry and contextual information (such as activity safety constraints) layers back into an interoperable twin representation. This addresses a critical gap in current construction robotics: the lack of integrated systems that simultaneously leverage pre-existing BIM data, real-time sensor streams, and autonomous decision-making for safe and efficient site navigation.

The framework fuses three data streams: (1) geometric and semantic information from BIM models as the foundational spatial and functional representation, (2) IoT/wearable activity data that describe work activity and safety exposure, and (3) UGV visual observations

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collected during site traversal. Accordingly, BIM2RDT implements an LLM-assisted orchestration layer as bounded semantic reasoning components that close the loop between sensing and model maintenance, representing an initial step towards agentic behavior, where LLMs are used iteratively via tool calls to transform observations into state updates and bounded actions, rather than only producing responses. As such, we use the term "agentic" in a narrow, operational sense where the AI module performs semantic reasoning that conditions perception (e.g., assigning gravity priors for ICP), triggers IFC-consistent twin updates, and proposes navigation constraints, without claiming autonomous long-horizon planning or full decision authority.

This study investigates three research questions: **RQ1:** How can semantic information from static BIMs be leveraged to constrain and improve point cloud registration accuracy in construction environments with ambiguous geometry and limited features? **RQ2:** To what extent can an LLM-based reasoning provide physically-informed, object-specific priors that reduce local-minimum failures in scan-to-BIM alignment? **RQ3:** How can sensor-detected safety activities be semantically mapped into interoperable digital twin representations to enable proactive intervention and context-aware robot planning?

To answer these questions, BIM2RDT implements a three-stage perception-to-update loop. BIM semantics are translated into context-specific prompts for open-vocabulary detection using YOLOE [4], enabling recognition of both permanent BIM families and temporary site objects without retraining. We introduce Semantic-Gravity ICP (SG-ICP), a point-cloud registration method where an LLM assigns object-adaptive upright priors per BIM family (e.g., heavy crates vs. easily tipped cones), producing physically plausible orientation regularization. Finally, sensor-detected activities are mapped into IFC constructs (*IfcEvent*, *IfcTask*), demonstrated via HAV monitoring against ISO 5349-1 thresholds [5] as a proof-of-concept. The resulting activity layers generate contextual cost maps for robot path planning. The specific contributions of this work are:

- A BIM-conditioned perception pipeline using BIM object labels to prompt open-vocabulary detection (YOLOE) and define ROIs for scan-to-BIM alignment.
- **SG-ICP**: a registration method using LLM-inferred, BIM-family-specific upright priors to regularize ICP and reduce local-minimum failures.
- An IFC-compatible mechanism to write sensor-recognized activities into the digital twin (via *IfcEvent*/*IfcTask*), demonstrated with HAV monitoring.
- Experimental validation showing RMSE reductions up to 97% in challenging scenarios and its consequences in context-aware path planning.

Fig. 1 shows an overview of the developed framework: robot sensing updates the twin through BIM-conditioned perception and SG-ICP registration; the updated twin then informs subsequent robot planning and safety-aware navigation

2. Literature review

This review is organized to mirror the BIM2RDT pipeline: (i) dynamic construction digital twins, (ii) robotic site monitoring and BIM-aided navigation, (iii) BIM semantics and ICP-based registration, and (iv) AI-based autonomous agents in construction.

2.1. From BIM to digital twin with geometric and operational updates

The evolution from Building Information Models (BIM) to digital twins (DTs) in construction represents a paradigm shift towards data-driven, real-time management of built environments. The defining characteristic that distinguishes a true DT from an advanced BIM model is this bi-directional, real-time or near-real-time data connection [6]. This cyber–physical integration establishes a continuous feedback loop: the physical asset is instrumented with sensors that feed data to the digital model, and the digital model, in turn, can be used to analyze, simulate, and control the physical asset. During construction, a dominant technical gap is *geometric* synchronization: maintaining an as-built geometry that stays aligned with BIM coordinates while temporary objects, staging, and work fronts evolve. This motivates scan-to-BIM workflows and automated update mechanisms that can operate continuously rather than as occasional manual reconciliation. Additionally, a construction digital twin requires an operational link between the physical site and its digital representation. Prior DT research has emphasized IoT-to-model connections for monitoring and analytics, often highlighting interoperability and real-time data streams as key enablers [6–8]. Wearables and IoT streams enable activity recognition and exposure monitoring, but these signals are often managed outside the BIM/DT representation, limiting their integration into planning and coordination. Hand-arm vibration (HAV) exposure, for example, is a representative safety signal because it has standardized evaluation metrics and actionable thresholds for wearable-based safety monitoring in construction settings (e.g., ISO 5349-1) [5,9].

2.2. Robotics reality capture and BIM-aided navigation

The continuous and accurate acquisition of as-built data is the lifeblood of a Digital Twin. While fixed sensors provide valuable data streams, mobile robotic platforms have emerged as a primary method for comprehensive and autonomous site scanning. Mobile robots provide repeatable sensing for progress and condition capture, reducing the cost and safety risk of manual scanning [10]. In indoor, GPS-denied construction environments, reliable autonomy requires robust localization and mapping under frequent changes and occlusions [11]. Prior work has explored UGV platforms for inspection and monitoring [12], and quadruped systems in particular offer superior traversal capability for unfinished interiors and cluttered work zones [13,14]. The fusion and registration of data collected from multiple scans and different sensor modalities (e.g., LiDAR, cameras) into a single, coherent model is another complex technical problem [15,16]. To address this, a recurring theme is leveraging building structure (planarity, corridors, cores) and pre-existing digital models to improve robustness with low computational cost. BIM-aided localization and mapping reduce drift and provide meaningful anchors, but success depends on accurate registration between robot observations and BIM pointclouds.

2.3. BIM semantics and geometric registration with ICP

The accurate alignment of 3D point clouds is a cornerstone of robotics, computer vision, and digital twinning for the built environment. The Iterative Closest Point (ICP) algorithm remains the foundational method for this task, but its limitations have inspired decades of innovation. The standard ICP algorithm iteratively refines the rigid transformation between two sets of point clouds by minimizing the sum of squared distances between corresponding points [17]. Despite its widespread adoption, ICP's performance is highly sensitive to the initial alignment and is famously susceptible to converging to local minima, especially in scenes with sparse features, repetitive geometry, or poor initial overlap [18]. In construction, robust point cloud registration (often referred to as "scan-to-BIM") is critical for as-built verification, progress monitoring, and digital twin maintenance. Researchers have extensively used ICP and its variants to align laser scans of a job site with the as-designed BIM model to detect deviations [19]. The structured nature of building environments, with prevalent planar surfaces, makes point-to-plane ICP particularly effective. Global BIM-point cloud registration minimizes deviations using ICP, evaluating alignments for construction quality control [20]. Lidar inertial odometry (LIO) and BIM have been used together for robust robot localization, using ICP-like methods to map and navigate in building environments [21].

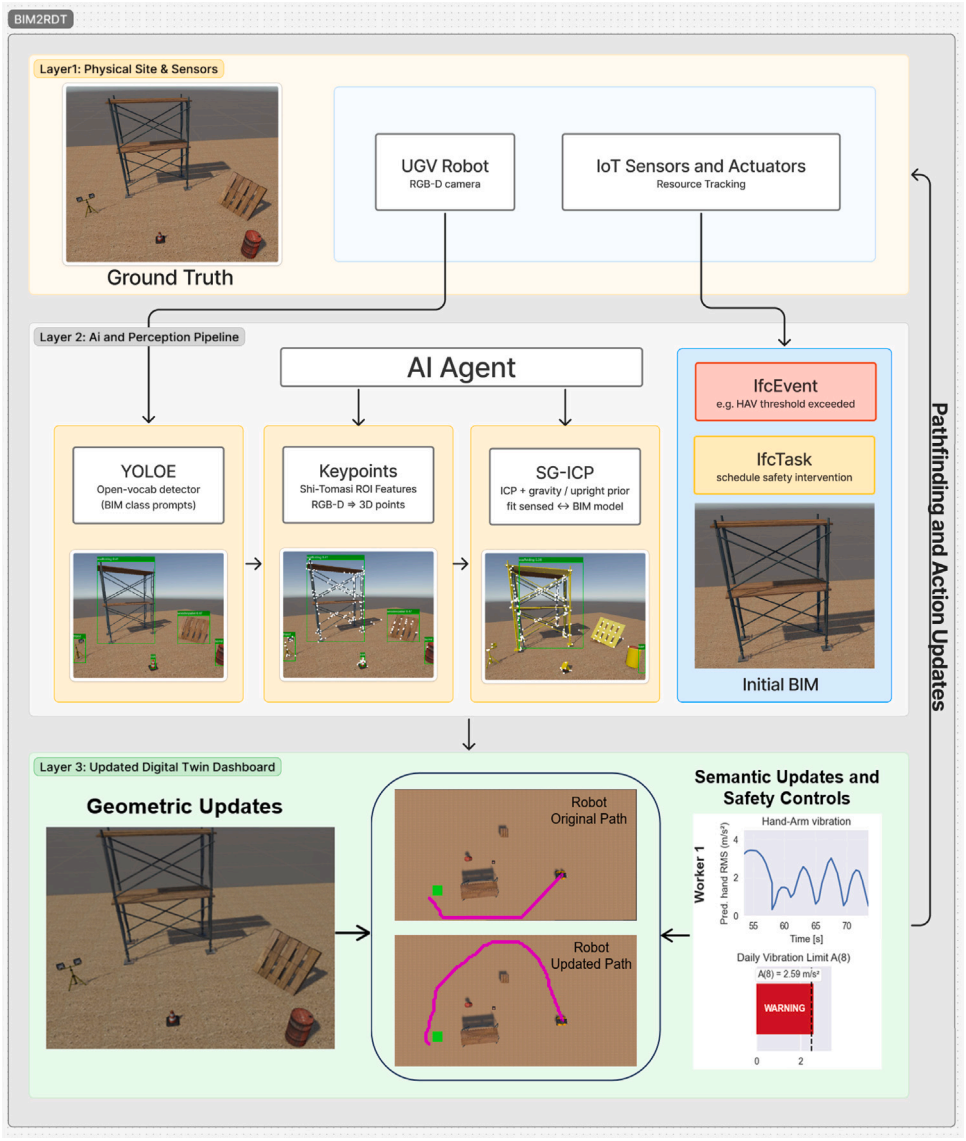


Fig. 1. Overview of the BIM2RDT framework.

Semantic-aided ICP in global LiDAR registration uses BIM semantics via Pose Hough Transform, enhancing robot autonomy in construction [22]. A common limitation is that gravity constraints are typically applied uniformly (a fixed upright prior), even though different site objects have different physical plausibility of tilt.

2.4. AI and autonomous agents in construction

Agentic AI systems are defined by capabilities such as proactive planning, contextual memory, sophisticated tool use, and the ability to adapt their behavior based on environmental feedback [23]. Unlike automation tools in the built-environment industry that follow rigid, pre-programmed instructions, or analytical models that simply provide outputs for human interpretation, agentic AI systems are designed to perceive their environment, reason about that information, and act autonomously to achieve specified goals [24]. This capacity for autonomous, goal-directed action makes them particularly well-suited for managing the unpredictable and dynamic nature of construction projects. Recent frameworks distinguish between simple AI agents (modular, task-specific systems) and true agentic AI, which exhibits multi-agent collaboration, dynamic task decomposition, persistent memory, and coordinated autonomy [25]. Industry practitioners have operationalized this concept through "agentic workflows" where AI systems iteratively reflect, plan, and use tools to accomplish complex tasks through multi-step processes rather than single-prompt responses [26]. In the construction context, this translates to systems capable of perception, planning, and action in unstructured environments, representing a critical evolution for managing construction execution complexity. Consequently, the development of intelligent agents capable of perception, planning, and action in unstructured construction environments represents a critical and underexplored research frontier [24].

2.5. Summary and gap addressed by BIM2RDT

Existing research provides strong building blocks such as robotic scanning, BIM-aided localization, ICP variants, and sensor-based safety monitoring. However, these are often treated as separate systems. BIM2RDT closes the loop by (i) conditioning perception on BIM semantics to stabilize object identification in dynamic inventories, (ii) improving scan-to-BIM alignment using SG-ICP with object-adaptive semantic-gravity priors, and (iii) writing safety-relevant activities into an interoperable twin representation that can directly inform robot planning. This structure directly motivates the methodology and matches the evaluation reported in the experiments.

3. Methodology

This section presents the BIM2RDT methodology in three stages. Sections 3.1–3.3 describe the perception pipeline, Section 3.4 introduces the core innovation of SG-ICP, and Sections 3.5–3.7 detail how safety activities and geometric updates are written back to the IFC model.

3.1. Object detection

We employ the YOLOE open-vocabulary object detector [4]. The selection of YOLOE is deliberate and critical to the BIM2RDT framework. Unlike traditional closed-set object detectors, which are limited to a predefined set of categories, open-vocabulary models can identify objects based on arbitrary textual descriptions. This capability is particularly suitable for construction sites, which are populated with a broad assortment and often variable set of object labels, from standard BIM components (e.g., columns, beams) to temporary materials (e.g., pallets, scaffolding) and unplanned obstacles. YOLOE allows our system to be highly adaptable, as it can be dynamically prompted with the specific BIM families relevant to the robot's current operational area without requiring model retraining, a significant advantage over alternatives.

Semantic information from the BIM or user-specified labels initializes a textual prompt that biases classification towards the target BIM families. The process begins by translating the semantic information from the BIM into a concise textual prompt for the detector. BIM family names, which can be verbose or contain non-descriptive metadata (e.g., ‘‘`Steel_I-Beam_W12x26_A992`’’), are processed using an LLM to extract the core object class. For example, the aforementioned name is simplified to "steel I-beam". Similarly, ‘‘`Traffic_Cone_Orange_70cm`’’ becomes "traffic cone". The prompt concatenates BIM family names, such as "Concrete Pillar" or "Safety Cone", with contextual descriptors like "construction site object" to bias classification. As an example, for a BIM family "Wall" the prompt might be: "a concrete wall in a building site".

This list of simplified class names, derived from the BIM elements expected in the scene, forms the textual prompt. For a given region, the prompt might be a list such as: ["steel I-beam", "traffic cone", "wooden crate", "steel barrel"]. This context-specific prompt biases the classification towards the target BIM families, improving detection accuracy and reducing false positives. The detector outputs bounding boxes with associated class names, which we then map one-to-one to the corresponding BIM families. These mappings define regions of interest (ROI) that support downstream feature selection and 3D registration analyses. In dynamic sites, this semantic biasing reduces false positives from clutter, ensuring ROIs align with BIM elements for accurate deviation detection.

3.2. Corner detector

In order to detect features within the 2D ROI, the shape geometry is required. We opt to use Shi-Tomasi corner detection to detect these features due to the algorithm's superior stability in tracking under affine transformations, which is beneficial for construction scenes with perspective distortions from robot viewpoints [27]. It is important to note that any corner detector algorithm can be used in this framework. The goal of the corner detector is to identify coordinates that exist on the object, and this set of 3D coordinates would give us plausible points that exist on the object geometry that can be used for point cloud alignment. Given a grayscale image I, compute spatial gradients I_x, I_y. For each pixel x in the ROI, form the second-moment matrix over a window $W(x)$:

$$M(x) = \sum_{u \in W(x)} \begin{bmatrix} I_x(u)^2 & I_x(u)I_y(u) \\ I_x(u)I_y(u) & I_y(u)^2 \end{bmatrix} \tag{1}$$

Let $\lambda_1(x) \geq \lambda_2(x)$ be the eigenvalues of $M(x)$. The Shi–Tomasi score is:

$$S(x) = \min\{\lambda_1(x), \lambda_2(x)\} \tag{2}$$

Accept x if $S(x) \geq \tau$ (cornerness threshold), then apply non-maximum suppression in a radius r, and keep the top K responses. The threshold τ is adaptively set based on image noise levels, typically 0.01–0.05, with $r = 3$ pixels and $K = 600$ for construction ROIs to balance density and computational efficiency. The Shi-Tomasi algorithm is a cornerstone in feature detection, known for its robustness and availability across OpenCV implementations. These detected corners within BIM-mapped ROIs provide robust keypoints for ICP correspondence. Since we are using a depth camera to provide perception to the robot, we can use the identified pixel (x, y) to query the z value so we get a 3D representation of the feature values that are detected on the 2D image. Given a set of points $A = \{(x_i, y_i, z_i)\}_{i=1}^{N}$ where $A \subset \mathbb{R}^3$, we can then conduct ICP with a root set of points $B = \{(x_i, y_i, z_i)\}_{i=1}^{N}$ where $B \subset \mathbb{R}^3$, which becomes our target points sampled from the BIM.

3.3. Pointcloud registration

In general BIM alignment approaches, the ICP algorithm aligns robot-captured point clouds (e.g., from LiDAR mounted on quadrupeds) with BIM-derived models, enabling deviation detection for progress monitoring [28]. The goal of using ICP is to minimize Wahba's problem [29], given by:

$$E_{\text{ICP}}(R, t) = \frac{1}{2} \sum_{i=1}^{N} w_i \parallel b_i - (Rr_i + t) \parallel_2^2 \tag{3}$$

While closed-form solvers such as the Kabsch algorithm can recover (R, t) given a set of *known* point correspondences [30], they require the data association to be specified *a priori*. However, the association is often not known in practice. To address this, ICP was introduced, which initializes data classifications given a defined heuristic. Concretely: ICP alternates between (i) estimating correspondences (nearest-neighbor matching under the current pose) and (ii) solving an optimization step for (R, t), until convergence or until a maximum number of iterations has been reached. This joint treatment allows ICP to handle unknown data association in practice.

However, ICP is known to converge to local minima. To mitigate this issue, many methods have been proposed. One of these methods is to introduce an additional source of information in the form of gravity alignment to promote a more upright final pose to improve point cloud alignment. However, the naive integration of bias can introduce error by dominating the point cloud registration optimization through excessively minimizing the gravity alignment term. This can be detrimental to BIM updates and potentially introduce discrepancies. This problem is addressed in BIM2RDT via semantic priors, and is explained in the following subsection.

3.4. Semantic-gravity iterative closest point

Consider the regularized optimization problem:

$$\min_{R \in \text{SO}(3), t \in \mathbb{R}^3} \frac{1}{2} \sum_{i=1}^{N} w_i \parallel b_i - (Rr_i + t) \parallel_2^2 + \frac{\gamma}{2} \parallel g_m - Rg_0 \parallel_2^2. \tag{4}$$

where $r_i \in \mathbb{R}^3$ are model sample points with normals, $b_i \in \mathbb{R}^3$ are corresponding scene points, $w_i \geq 0$ are robust weights, $(R, t) \in \text{SO}(3) \times \mathbb{R}^3$ is the unknown rigid transform, g_0 is the canonical upright vector in model coordinates (e.g. e_z), g_m is the measured gravity vector in world coordinates, and $\gamma \geq 0$ is the gravity weight, which balances data fidelity and upright bias; in practice, we tune it via cross-validation on simulated construction scenes.

It is important to note that this regularization is algorithm agnostic. Given any point cloud registration and data association scheme, one

can introduce a gravity bias regularization term to introduce upright behavior with a tunable weight in γ. In our experiments, we experiment with commonly used algorithms to show their efficacy in BIM settings by leveraging the semantic qualities.

3.4.1. Gravity alignment

The data term

$$E_{\text{ICP}}(R,t) = \frac{1}{2}\sum_{i=1}^{N} w_i \,\| b_i - (Rr_i + t)\|_2^2 \tag{5}$$

is a standard weighted ICP.

The gravity bias term

$$E_{\text{grav}}(R) = \frac{\gamma}{2}\,\| g_m - Rg_0\|_2^2 \tag{6}$$

acts as a soft constraint, encouraging Rg_0 to align with gravity, maintaining a more upright orientation. Expanding the gravity penalty:

$$E_{\text{grav}}(R) = \frac{\gamma}{2}\left(\|g_m\|_2^2 + \|Rg_0\|_2^2 - 2\,g_m^{\top} Rg_0\right) \tag{7}$$

Since $\|g_m\| = \|g_0\| = 1$ in unit-norm form, this reduces to

$$E_{\text{grav}}(R) = \gamma\left(1 - g_m^{\top} Rg_0\right) \tag{8}$$

Thus, minimizing E_{grav} is equivalent to maximizing the alignment of Rg_0 with g_m.

In Gauss–Newton or SVD updates, this bias is equivalent to adding one weighted correspondence (g_0, g_m) with weight $\sqrt{\gamma}$:

$$\{(r_i, b_i, \sqrt{w_i})\}_{i=1}^{N} \cup (g_0, g_m, \sqrt{\gamma}) \tag{9}$$

3.4.2. Semantic alignment

We use BIM semantic labels to estimate each object's susceptibility to gravity-induced tipping. Heavier objects tend to have a stronger upright bias; if they appear tilted, routine site dynamics or workspace drift are less likely explanations. This extends prior pose-informed ICP (e.g., IMU-ICP [31]) by replacing sensor-derived pose bias with an explicit upright vector (e.g., the gravity vector) and by making the bias object-adaptive through BIM semantics. This approach helps in reducing errors in tilted but stable objects like heavy equipment. We define

$$\gamma_{new} = \gamma_{family} \cdot \gamma_{initial} \tag{10}$$

where $\gamma_{initial}$ is the initialized value that can serve as a general tuning knob. Assigning weights manually is cumbersome, so we instead use an LLM to traverse all BIM families and assign $\gamma_{\text{family}} \in [0, 1]$, a prior for the object being upright. The LLM interprets family names, descriptions, and construction-site context and outputs the corresponding bias score. We use the OpenAI API to prompt GPT-4o-mini. Specifically, the LLM is given the following context:

> "You return BIM alignment priors. For each label, output: upright bias $\in$ $[0, 1]$ (fraction of tilt corrected per ICP iteration). Only include labels given. No extra text. Take gravity into account, how likely is this to not be upright? If it is not likely we need a smaller value. JSON that matches the provided schema strictly."

A JSON file containing the name of all families is attached to the command. The LLM will use its contextual reasoning capabilities to infer the possibilities of being "upright" and returns a tailored coefficient. Algorithm 1 describes the SG-ICP registration procedure, which augments standard ICP with a gravity-alignment bias and object-dependent semantic weighting. Given an observed point cloud and BIM-derived priors, it iteratively estimates the rigid transformation that aligns detected objects to the model while favoring physically plausible (upright) solutions.

Algorithm 1 SG-ICP pseudo code incorporating gravity bias

Require: body B; loss $\mathcal{L}$, schedule L, default bias β_{def}; gravity $\hat{g}$; params: *restarts*, *yawJitterDeg*, *sampleCount*, *maxIterations*, *maxPairDistance*, *normalGate*, *trimFraction*, *huberDelta*, $(\varepsilon_r, \varepsilon_t)$
Ensure: pose of B aligned to scene S

```
(β, yawOnly) ← LLMBiasManager(ℒ, L, β_def)
S ← AcquireScenePoints()    require |S| ≥ 6
(V, N) ← SampleModelWithNormals(B, sampleCount)
(p_0, q_0) ← SnapshotPose(B); PrepareMovementMode(B)
for a ∈ YawRestarts(restarts, ĝ, yawJitterDeg) do
    ResetPose(B, p_0, q_0); ApplyYawJitter(B, a)
    for k = 1..maxIterations do
        (X, N_W) ← ModelToWorld(B, V, N)
        P ← MatchPairs(X, S, maxPairDistance, N_W, normalGate)
        P ← RelaxIfFew(P, S, X);    P ← TrimPairs(P, trimFraction)
        w ← RobustWeights(P, huberDelta)
        (R_Δ, t_Δ) ← SolveRigid(P, w, β)
        R_Δ ← ProjectYawIf(R_Δ, ĝ, yawOnly)
        ApplyDelta(B, R_Δ, t_Δ)
        if Converged(P, R_Δ, t_Δ, ε_r, ε_t) then break
        end if
    end for
    ScoreAndKeepBest(B, S, maxPairDistance)
end for
```

3.5. Safety-aware digital twin updates with activity integration

Once the robot-captured point cloud is geometrically registered to the BIM using the Semantic-Gravity ICP (SG-ICP) method, the BIM2RDT framework extends these capabilities to incorporate real-time activity recognition for enhanced safety monitoring. Here, it semantically links sensor-detected activities to BIM entities using IFC standards. Specifically, it leverages *IfcTask* and *IfcEvent* (core entities in the IFC schema for representing identifiable units of work or activities) as a unifying interface, allowing sensor-derived activities (e.g., "hammer drilling with unsafe vibration levels") to be treated as compatible IFC constructs without requiring explicit temporal scheduling ties.

The integration of dynamic site conditions beyond geometry is exemplified by the analysis of worker activities involving hazards such as excessive hand-arm vibrations (HAV) during tool operation. We draw from prior work on wearable sensor-based HAV monitoring [9], which demonstrated the feasibility of attaching wearables to positions beyond those specified in ISO 5349-1 [5]. In this work, we expand upon the existing methodology and develop a subject-specific approach that enables more accurate mapping.

The acceleration signals are transmitted to a central processing unit, where the frequency-weighted RMS acceleration a_{hv} is calculated. This data serves as the input for computing *A(8)*, the 8-h energy-equivalent frequency-weighted vibration total value, as defined in [5]. The initiation of the activity is marked by the creation of an *IfcEvent*. Upon attaining the critical exposure action value (EAV) of 2.5 m/s^2, a triggering mechanism schedules an *IfcTask*, thereby assigning a safety instructor to intervene and mitigate the vibration-induced workload on the affected worker. Following the termination of the activity, another *IfcEvent* containing the accumulated daily vibration exposure is created, enabling the precise tracking and verification of compliance with ISO 5349-1 limits. On the backend, we enrich the existing IFC file with time-stamped HAV events using IfcOpenShell.

In this work, HAV detection serves two roles: (i) a long-horizon occupational exposure signal that is logged and traced in BIM/IFC to support safety management workflows and (ii) a short-horizon risk signal that enriches information used by robot navigation to reduce traversal through high-exposure work zones. Sensor-recognized activities are written into the twin as Activity Instances that can reuse the

project's schedule vocabulary (labels, scopes) when present, but remain valid even when no schedule file, look-ahead, or CPM alignment exists. This mapping exploits IFC's extensible structure, where sensor activities are instantiated as new or augmented entities with attributes for safety metrics (e.g. a_{hv} values and risk flags), ensuring interoperability without altering the core BIM model.

Work planning is not typically part of baseline BIM/IFC deliveries, so representing it using *IfcTask* and *IfcEvent* requires explicit modeling preconditions. BIM2RDT therefore treats *IfcTask/IfcEvent* as an optional runtime enrichment layer that can operate in two modes. If an *IfcWorkSchedule* structure already exists (e.g., from 4D planning), BIM2RDT appends HAV-related *IfcEvent* instances and links them to the corresponding existing *IfcTask*. If no work plan is present, BIM2RDT creates a minimal process scaffold for traceability only: *IfcEvent* instances record observed activity start/end and threshold exceedances, and a minimal *IfcTask* encodes a recommended intervention (e.g., "Stop worker from HAV exposure"). This schedule-agnostic mode is not intended to reconstruct project scheduling, yet it provides a consistent, queryable record of safety-relevant observations.

To support actionable alarms and queries, each event must be grounded in a work context. Specifically, the spatial grounding of each *IfcEvent* is achieved through the BIM object that represents the vibration source. When the HAV monitor identifies a threshold exceedance, the system uses the LLM-based semantic mapping (Section 4.7.1) to associate the event with the most likely source object in the IFC model. The *IfcLocalPlacement* of that object provides the geographic reference for the event, and the subsequent *IfcTask* references both the affected worker (identified by sensor assignment) and the spatial zone defined by the object's placement. Real-time alarms are triggered by the HAV threshold logic and routed through a web-hosted dashboard, where the event location, worker identity, vibration magnitude, and recommended intervention are displayed together.

3.6. Activity-proximity cost map

In our method, HAV is associated with a workstation location rather than continuous worker tracking. The detection of HAV is assigned to a BIM object representing the vibrating source. The spatial placement of this BIM object provides the event location. SG-ICP provides an accurate geometric map, and the HAV monitor augments it with contextual activity layers. When a high HAV activity is recognized, the LLM agent infers the most likely activity family and instantiates a local cost map around that object. For a grid cell at (x, y) and an object reference point $o = (x_o, y_o)$, the activity cost decays with distance from o as:

$$f(x, y) = A\, e^{-d(x,y)} \tag{11}$$

where $A > 0$ is a design parameter (the peak cost at the object) and $d(x, y) \geq 0$ is the distance from (x, y) to o. Unless noted otherwise, we take

$$d(x, y) = \left\| (x - x_o,\ y - y_o) \right\|_2 \tag{12}$$

but any other distance metric would be valid.

While this produces a costmap that can both qualitatively and quantitatively be beneficial for project managers to measure the effects and probabilistic areas of HAV activity, it can also be included in robot mission planning. Previous works have shown that natural language sentiment analysis can directly be used in classical robot path planning [16]. We later on demonstrated the efficacy of using BIM2RDT in path planning scenarios.

3.7. Automated digital twin synchronization via IFC updates

Once SG-ICP computes object transformations in robot coordinates, these updates must be written back to the BIM to synchronize the digital twin with observed site conditions. This process involves two stages: semantic object mapping and coordinate system transformation.

3.7.1. Semantic object mapping

After SG-ICP computes alignment transformations, the system must map detected objects to their corresponding IFC elements. YOLOE (Section 3.1) outputs object labels. IFC elements are referenced by globally unique identifiers (GUIDs). An LLM bridges this semantic gap by mapping YOLOE-detected object names to IFC element GUIDs using reasoning over object types and descriptions. The following prompt is used:

> "You are tasked with mapping detected construction site objects to IFC building elements based on their names and types.
>
> Detected Objects (from YOLOE):
>
> - Wooden Crate (hierarchy: Environment/Props/WoodenCrate) - ...
>
> IFC Objects (from BIM):
>
> - Crate_Storage_01 (GUID: 2a3b4c5d..., Type: IfcFurniture) - ...
>
> Rules: 1. Only map objects where you are confident there is a match 2. Use exact names and GUIDs as provided 3. Consider semantic similarity 4. If uncertain, set confidence to 'low' and explain why 5. Return valid JSON"

The LLM returns structured JSON mappings with confidence scores ("high", "medium", "low") and semantic reasoning for each match. A rule-based fallback provides limited pattern matching when API access is unavailable, however, this fallback only succeeds for highly standardized naming and cannot handle the semantic variability common in construction sites.

3.7.2. Coordinate system transformation

SG-ICP outputs are converted from robot coordinate frames to IFC coordinate conventions and applied to update placements of the LLM-mapped objects in the IFC model. The rigid transformation resulting from the SG-ICP registration represents the alignment of the physical object relative to the digital model. To synchronize the Digital Twin, this transformation is first converted from the robot's sensing coordinate frame to the IFC project's coordinate convention (addressing axis orientation and unit scaling differences). We utilize the *IfcOpenShell* library to programmatically access and modify the IFC schema. Using the unique element GUID identified in Section 3.7.1, the system targets the specific *IfcProduct* and updates its *IfcLocalPlacement*. The computed translation and rotation matrices are applied to the object's *IfcAxis2Placement3D* attributes, ensuring that the geometric state of the digital asset is persistently updated to match the physical site conditions observed by the robot.

4. Experiments and results

This section presents experimental validation of the BIM2RDT framework across four key components: (1) Semantic-Gravity ICP alignment accuracy under varying viewpoints and noise conditions, (2) integration with RANSAC-based registration, (3) safety monitoring through HAV activity mapping, and (4) context-aware path planning. Experiments are conducted in controlled simulation environments designed to isolate and evaluate individual framework components, establishing baseline performance and demonstrating proof-of-concept for each module. While these controlled scenarios enable repeatable measurement and systematic ablation studies, field deployment in active construction sites remains as future work (see Section 6.2). Each scenario configuration is evaluated in a single trial with all stochastic elements controlled by fixed random seeds.

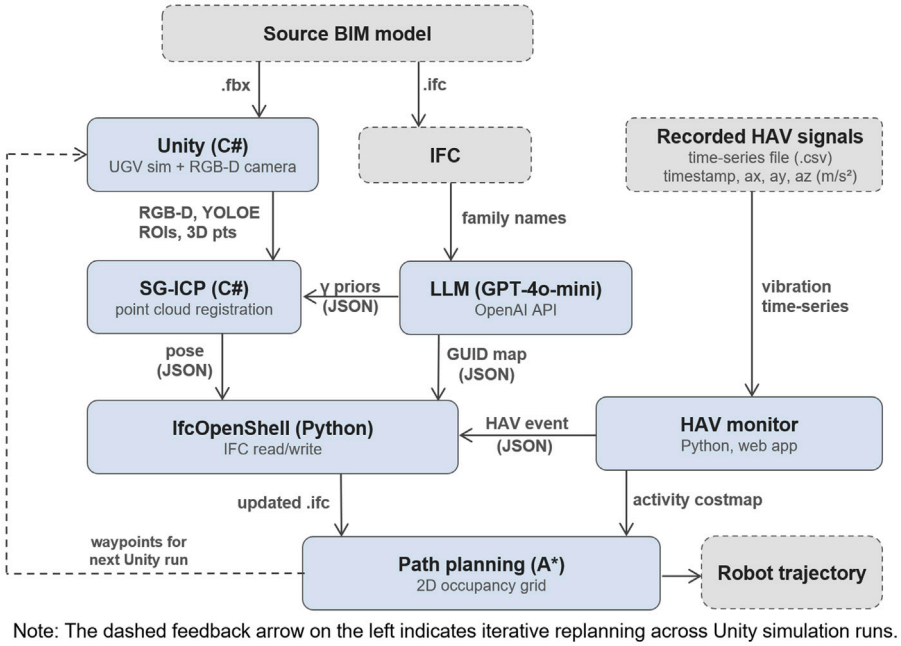


Fig. 2. Implementation environment of the BIM2RDT pipeline showing software modules (blue, solid border), external data and outputs (gray, dashed border), and the data formats exchanged at each interface.

4.1. Implementation environment

Fig. 2 complements the conceptual overview in Fig. 1 by providing an implementation-level schematic that traces each data handover, including file formats, interchange protocols, and software interfaces for reproducability purposes, through the pipeline as built for the experiments reported below.

All experiments were conducted in a controlled simulation environment built in Unity. The environment models a virtual UGV equipped with a simulated RGB-D camera capturing color and depth images at configurable viewpoints. BIM objects are represented as textured 3D meshes exported from the source BIM model authored in Revit, as FBX files for 3D rendering and IFC files for semantic processing, with recorded ground-truth poses for quantitative evaluation. HAV sensor signals were pre-recorded from physical wearable devices using the sensor configuration and placement protocol described in [9] and stored as CSV time-series files (columns: timestamp, tri-axial acceleration), then replayed in simulation to drive the safety monitoring pipeline.

The BIM2RDT pipeline is implemented primarily in C# within Unity, with Python-based components for IFC manipulation and safety monitoring. Object detection uses YOLOE with BIM-derived textual prompts; feature extraction within detected ROIs uses the Shi-Tomasi corner detector via OpenCV version 4.13.0. Point cloud registration, including the SG-ICP extension, is built in Unity as well, with RANSAC-based global initialization. ICP was run with a maximum of 90 iterations and was early-stopped when the threshold of 0.0001 was reached. Semantic reasoning for gravity-prior assignment and object-to-IFC mapping is performed through the OpenAI API (GPT-4o-mini), with outputs parsed as schema-validated JSON, specifically, BIM-family-specific gravity priors γ and detected-object-to-IFC-GUID mappings, both exchanged as structured JSON between the LLM and the registration and twin-update modules.

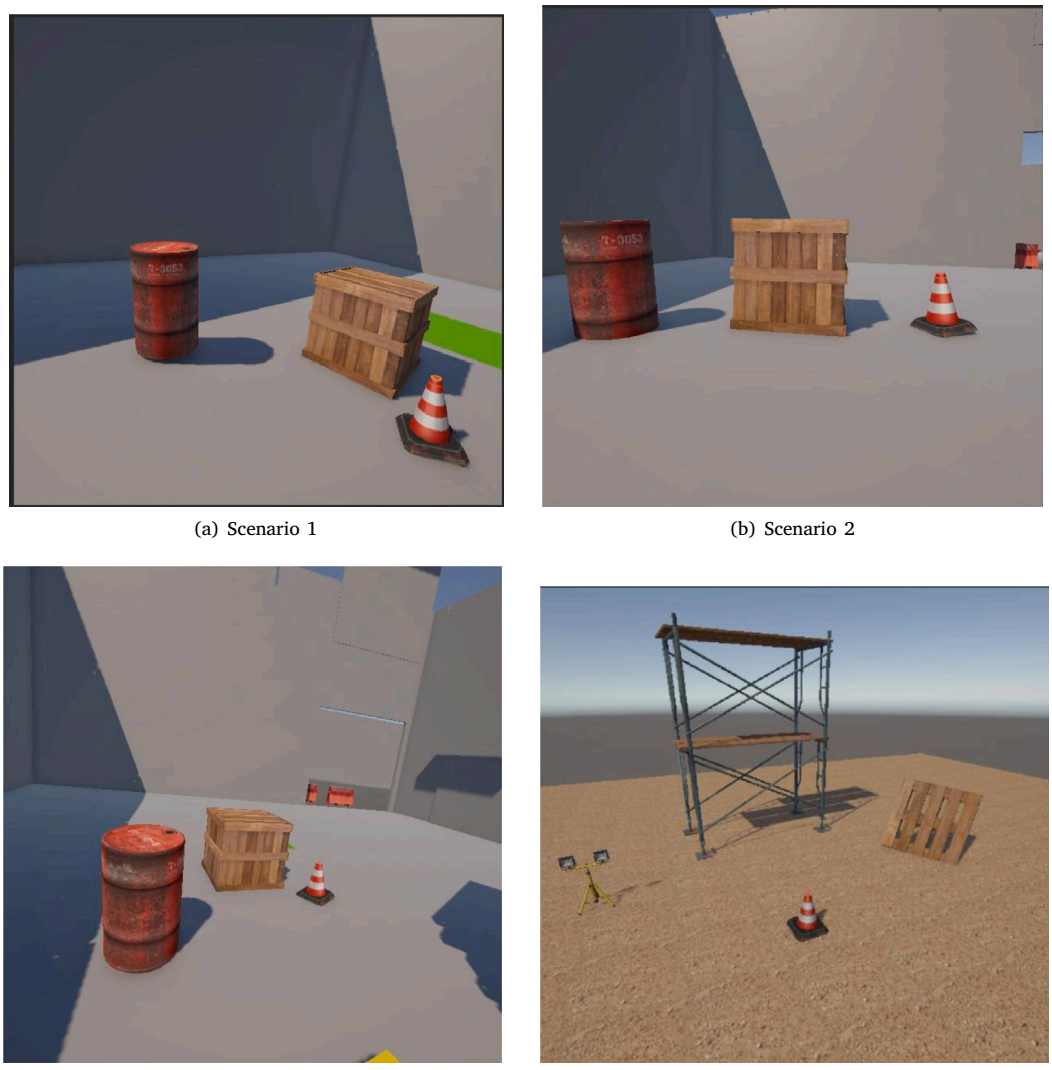
Digital twin updates are written to IFC files using IfcOpenShell version 0.8.4.post1, which programmatically modifies IfcLocalPlacement attributes and instantiates IfcEvent/IfcTask entities for HAV-related safety records. The HAV monitoring dashboard (Fig. 5) is implemented as a web application using Python 3.12 with real-time vibration traces and threshold alerts. Path planning (Fig. 7) operates on a 2D discretized occupancy grid derived from the updated BIM floor plan, with A* search over obstacle cells and activity-proximity cost overlays. All experiments were conducted on a system with an AMD Ryzen 5 3600 6-Core Processor, 16 GB of RAM, and an NVIDIA GeForce RTX 2060 Super. Each scenario configuration uses fixed random seeds to ensure reproducibility. Furthermore, to support full reproducibility, the source code for all pipeline components is available from the corresponding author upon request.

4.2. Semantic-gravity ICP (SG-ICP)

Here, we give mesh results on how well the meshes overlap, given the ground truth and SG-ICP. We experiment in four different scenarios. One is the same setup with three different viewpoints, and the fourth one is a variety of BIM families, which would have poor feature description (scaffolding, tripod, wooden pallets, etc.). We experiment with various perspectives to show the robustness under different feature sets. Both ICP implementations use Huber weights, restart poses, and randomize initial conditions to further improve convergence. The four scenarios can be seen in Fig. 3. Each scenario contains 3-5 BIM objects with varying geometric features. Scenario 1-3 feature simple primitives (cones, barrels, crates) and Scenario 4 introduces complex scaffolding geometry, representing challenging construction site conditions.

4.3. Random sample consensus

We obtain an initial rigid alignment using a RANdom SAmple Consensus (RANSAC)-based global registration stage [32]. Feature correspondences are generated between the downsampled source and target point clouds, and RANSAC is used to robustly estimate an initial transform T_0 by maximizing the number of inliers under a distance threshold τ. Starting from T_0, we perform local refinement using ICP to minimize the objective function. Since RANSAC is used commonly as a method to filter out incorrect correspondences before conducting ICP, we can still incorporate the gravity residual inside the optimization. This emphasizes the plug-and-play feature of our approach.



(a) Scenario 1 (b) Scenario 2

(c) Scenario 3 (d) Scenario 4

Fig. 3. Four evaluation scenarios used for the mesh-overlap analysis.

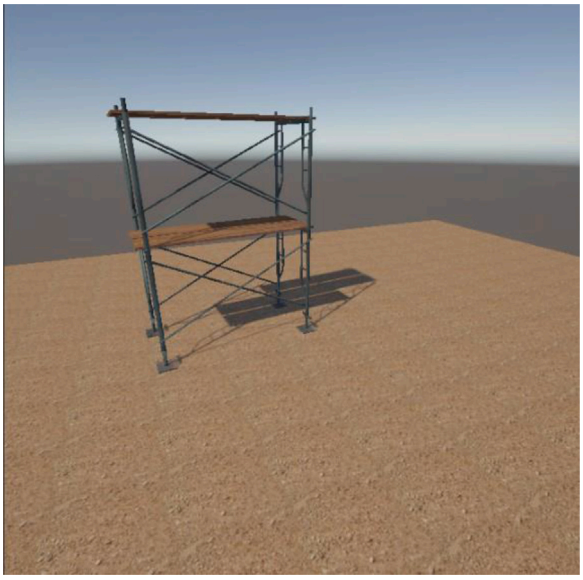


Fig. 4. Scenario for noisy conditions.

4.3.1. Ablation studies

We conduct ablation studies to isolate the impact of the gravity term and the semantic (family-specific) weighting used in SG-ICP. Using the four multi-object scenarios in Fig. 3, we evaluate six variants: (i) ICP, (ii) ICP + gravity regularization with a single static weight (static γ), (iii) SG-ICP (family-weighted γ), and the same three variants preceded by a RANSAC-based global initialization (RANSAC, RANSAC + static γ, RANSAC + SG-ICP). Each configuration is evaluated on the same scene with fixed random seeds to ensure reproducibility. Alignment quality is reported as Hausdorff-distance RMSE between the aligned mesh and the ground-truth pose; Table 1 summarizes these RMSE results across all four scenarios.

To stress-test robustness to sensor noise, we additionally evaluate single-object alignment under additive Gaussian depth noise, using the noisy-condition setup in Fig. 4. Table 2 reports the resulting normalized RMSE under noise for each variant. At moderate noise ($\sigma = 0.1$), SG-ICP reduces RMSE by 56% relative to standard ICP (3.04 vs. 6.93), while at high noise ($\sigma = 0.2$), the reduction is 97% (2.50 vs. 88.73). The RANSAC-initialized variants show a similar pattern: RANSAC SG-ICP achieves 1.93 at $\sigma = 0.1$ compared to 42.61 for standard RANSAC, a 95% reduction. These results confirm that semantic-gravity priors provide the greatest benefit precisely when alignment conditions are most challenging.



Fig. 5. Integration of HAV monitoring into the updated digital twin, with automatic warning at the daily vibration limit.

Table 1
Hausdorff distances (RMSE).

Method (RMSE)	Scen. 1	Scen. 2	Scen. 3	Scen. 4
ICP	4.964481	10.943896	4.111695	**54.700146**
ICP (static γ)(ours)	5.511244	4.895416	3.870467	59.895943
SG-ICP (ours)	**4.910435**	**3.964882**	**3.632635**	57.057274
RANSAC	5.435321	5.719707	4.976137	**30.579460**
RANSAC (static γ)(ours)	4.952999	**4.325600**	4.072346	54.715191
RANSAC SG-ICP(ours)	**4.769428**	5.496947	**3.609692**	35.954880

Table 2
Normalized Hausdorff Distance RMSE under different noise levels.

Method	$\sigma = 0.1$	$\sigma = 0.2$
RANSAC_SG-ICP	**1.926605**	43.968082
RANSAC_Fixed (static γ)	3.510232	**43.646484**
RANSAC	42.609661	99.852318
SG-ICP	**3.036388**	**2.501471**
ICP_Fixed (static γ)	3.073769	3.680148
ICP	6.930099	88.734306

4.4. Hand-arm vibration activity mapped in the DT

As shown in Fig. 5, the HAV dashboard systematically records and processes vibration data, subsequently generating visual warnings upon violation of the EAV. The dashboard is designed as a web application to be easily reproducible and scalable for real-world applications. Simultaneously, the *IfcEvent* start, end, accumulated daily vibration and the subsequent *IfcTask* of intervention are written to the IFC file. Fig. 6 shows a minimal representation of an IFC file, containing the concatenated events. IFC files serve as an industry-wide accepted format, yet it must be noted that they cannot be displayed easily due to its size. Its sequential structure allows adding elements without disturbing the original version. For readability, some elements have been removed. In addition to *IFcEvent* and *IFcTask*, the helper elements *IfcWorkSchedule* and *IfcDeclares* are shown. For visualization purposes only, the example replay accelerates the time axis and the recorded event ordering and threshold logic are unchanged.

4.5. Bim2RDT-enabled path planning

In this section, we demonstrate the efficacy of using BIM2RDT for context-aware path planning against traditional BIM path planning. While BIM-based path planning has been thoroughly investigated and validated, most works rarely acknowledge the safety considerations within the BIM. Here, we evaluate three sequential scenarios. In the first scenario, the BIM is used naively for path planning with A*. However, the BIM is outdated and does not accurately represent the real world. In the second scenario, we apply the first stage of our pipeline, SG-ICP, to update and align the BIM, enabling more accurate path planning. In the third scenario, we introduce an HAV activity event, where the BIM2RDT method assigns avoidance costs based on the event context.

Table 3
Path metrics comparison.

Metric	BIM2RDT	SG-ICP Only	Native BIM A*
Path length (units)	17.73	14.45	7.50
Min-obstacle distance (m)	1.390	0.139	0.000
Avg-obstacle distance (m)	2.648	2.265	N/A
Collision	No	No	Yes

Fig. 7(a) shows the overall real-world setup and Fig. 7(b) illustrates the overlap between the BIM (highlighted in yellow) and the real-world environment (rendered with real textures). A significant misalignment is apparent, which can lead to incorrect path planning when the BIM is used directly. After a single iteration of SG-ICP, the alignment between the BIM and the real-world scene improves substantially, as shown in Fig. 7(c). Fig. 7(d) shows the resulting A* path before the SG-ICP correction, which results in a collision with a real-world object, marked by an "X" symbol. However, planning with A* on the updated BIM produces a collision-free path, illustrated in Fig. 7(e). Finally, when BIM2RDT is fully enabled, additional safety constraints, such as HAV, are incorporated. This results in the path shown in Fig. 7(f), which appropriately avoids cluttered regions of the workspace.

Table 3 quantitatively supports these observations. In the first scenario, which relies solely on A*, the planned path collides with a real-world obstacle. In the second scenario, following SG-ICP alignment, the planner successfully reaches the goal; however, this stage does not account for safety or contextual constraints. When BIM2RDT is activated, the planner generates a successful and conservative path that safely connects the start and goal locations.

The path length increase from A* (7.5 units) → SG-ICP (14.45) → BIM2RDT (17.73) reflects progressive safety layering. SG-ICP ensures geometric feasibility (no collisions), while BIM2RDT adds HAV avoidance, increasing clearance from the workstation by 2.3 m. This 23% path length penalty is typical for human-aware motion planning, where safety buffers necessarily reduce efficiency.

```
ISO-10303-21;
HEADER;
FILE_DESCRIPTION(('ViewDefinition [CoordinationView, ...);
FILE_NAME('Project_BIM2RDTifc','2025-09-19T00:02:56-08:00',...);
FILE_SCHEMA(('IFC4X3_ADD2'));
ENDSEC;
DATA;

...

#33915=IFCWORKSCHEDULE('genID_1',$,'HAV Schedule 2025-09-19','HAV activity log and safety
warnings', ...);

#33916=IFCRELDECLARES('genID_2',$,$,$,#38,(#33915));

#33917=IFCEVENT('genID_3',$,'HAV Activity Start','timestamp=2025-09-
19T20:35:19Z',$,$,$,.STARTEVENT.,.USERDEFINED.,'HAV_ACTIVITY_START',$);

#33919=IFCEVENT('genID_5',$,'HAV Daily Limit Reached (Worker 1)','timestamp=2025-09-19T20:35:38Z;
worker=1; A8=2.506 m/s2',$,$,$,.INTERMEDIATEEVENT.,.USERDEFINED.,'HAV_DAILY_LIMIT_REACHED',$);

#33921=IFCTASK('genID_7',$,'Stop Worker 1','Stop HAV exposure due to threshold
exceedance',$,$,$,$,'Stop HAV exposure',.F.,$,$,.USERDEFINED.);

#33923=IFCRELSEQUENCE('genID_9',$,'HAV Threshold \X2\2192\X0\ Stop Task','Trigger stop action when
HAV threshold reached',#33919,#33921,$,.FINISH_START.,$);

#33924=IFCEVENT('genID_10',$,'HAV Activity End (Worker 1)','timestamp=2025-09-19T20:35:39Z;
worker=1; A8=2.543 m/s2',$,$,$,.ENDEVENT.,.USERDEFINED.,'HAV_ACTIVITY_END',$);


ENDSEC;
END-ISO-10303-21;
```

Fig. 6. HAV events are written to the IFC file.

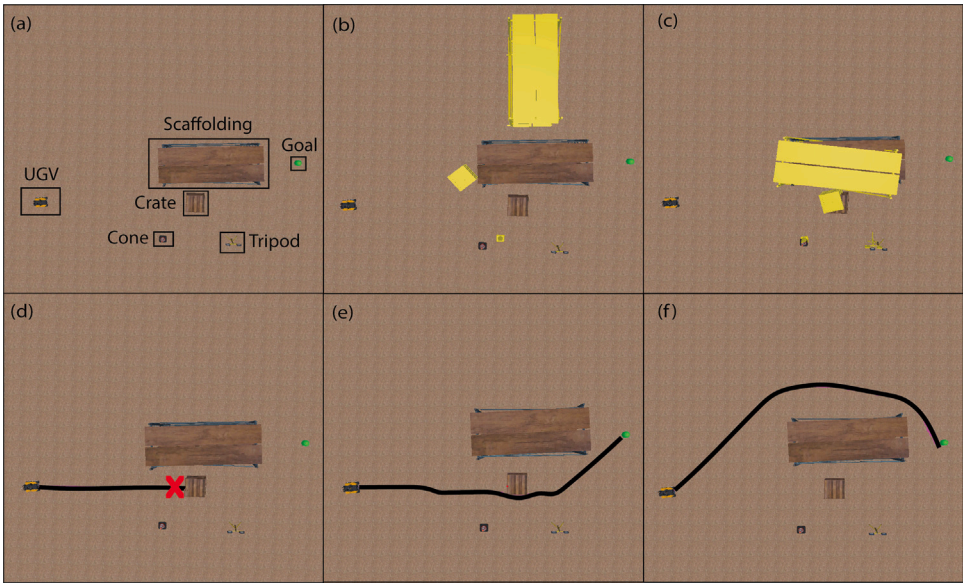


Fig. 7. Path planning results by using BIM2RDT. (a) Real-world scene layout with labeled objects; (b) BIM overlay (yellow) showing misalignment with real-world positions; (c) SG-ICP-corrected alignment; (d) Naive A* path; (e) A* path after SG-ICP correction; (f) BIM2RDT safety-aware path avoiding HAV activity zone.

4.6. Digital twin update via automated IFC synchronization

To demonstrate the end-to-end pipeline, the complete BIM2RDT workflow was applied to the scene shown in Fig. 8. Three objects (i.e., a wooden crate, a traffic cone, and a barrel) were detected by YOLOE from the UGV's RGB-D feed. SG-ICP was then used to align each object's extracted point cloud with the corresponding BIM-derived geometry. Following alignment, the LLM-based semantic mapping module (Section 3.7.1) was invoked to associate each detected object with its IFC counterpart. Table 4 shows the results. The Confidence score is a categorical assessment ('high,' 'medium,' or 'low') produced by the LLM as part of its structured JSON output, based on the degree of semantic similarity between the detected object label and the IFC element's name and type attributes. A 'high' confidence indicates a direct, unambiguous match (e.g., 'Cone' to 'Traffic_cone:750 mm'), while 'medium' reflects cases requiring inference across differing naming conventions (e.g., 'WoodenCrate' to 'Box:33x38x30'). Following successful mapping, the SG-ICP rigid transformations were converted from the robot coordinate frame to IFC conventions and applied to update each element's *IfcLocalPlacement*. Fig. 8 shows the resulting updated IFC

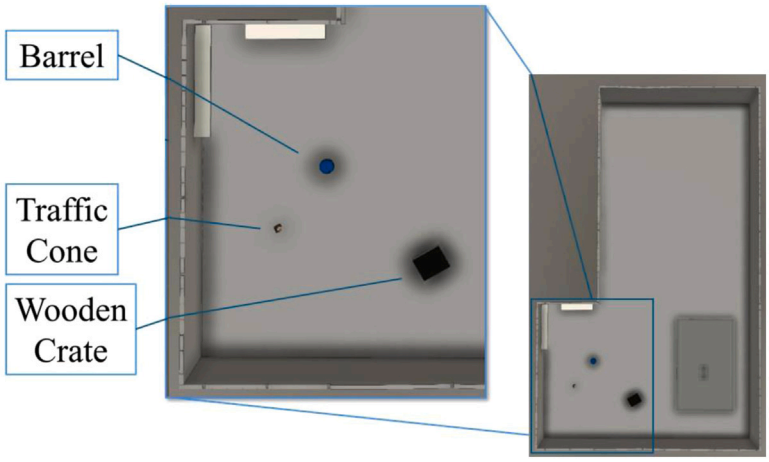


Fig. 8. Updated IFC file geometry after automated pipeline.

Table 4
LLM semantic mapping results.

Detected object	IFC element	IFC type	Confidence
WoodenCrate	Box:33 × 38 × 30	IfcFurniture	Medium
Cone	Traffic_cone:750mm	IfcBuildingElementProxy	High
Barrel	Barrel:55 US gallon	IfcBuildingElementProxy	High

model geometry. Geometry extracted and rendered directly from the IFC file using IfcOpenShell.

Following successful semantic mapping, the SG-ICP transformations were converted from robot coordinates to IFC conventions and written to the IFC model.

This demonstration validates the BIM2RDT methodology, as LLM-based semantic reasoning successfully maps detected objects to IFC elements, and the updated IFC file accurately reflects the robot-observed site conditions.

5. Discussion

In this section, the obtained results are discussed from both theory and practice standpoints.

5.1. Implications of geometric alignment results

The experimental outcomes for SG-ICP highlight its superior performance in aligning robot-captured meshes with BIM models, as evidenced by the substantial reductions in Hausdorff distances across diverse viewpoints. Notably, the high improvements in the first three scenarios demonstrate how semantic-gravity integration addresses common ICP pitfalls, such as sensitivity to initial poses in environments with ambiguous features, analogous to construction sites with often occluded elements like scaffolding or equipment stacks.

This robustness enhances the reliability of digital twins for as-built verification, potentially minimizing costly rework by enabling early detection of geometric deviations. There can be occurrences where ICP does better than the gravity-aligned approaches, such as the last scenario. This is expected since the gravity alignment introduces different rotations and translations, since it can sacrifice the optimal point cloud transformation for a more upright pose. If the correspondences are of high quality, this gravity bias introduces higher errors in the optimization of the true pose. However, this upright bias serves as a compass in noisy environments where the correspondences are poor, which can guide more realistic orientations as opposed to the best pose under poor pointclouds.

Furthermore, the mixed magnitude of improvements across scenarios with static scaling argues for adaptive priors. Specifically, schedule and scene context (when available) or learned scene cues can modulate the semantic-gravity weight, preserving the strong performance in difficult cases without over-constraining easy ones. This aligns with our methodology goal of LLM-mediated parametrization, "common sense" informs the bias where helpful and abstains where unnecessary.

5.2. Practical implementation of BIM2RDT in construction settings

The successful deployment of BIM2RDT within active construction projects requires careful consideration of how this methodology integrates with established project delivery processes, existing technology infrastructure, and field operations management. Construction practitioners face unique challenges when implementing advanced digital twin technologies that extend beyond pure technical functionality to encompass organizational change management, workflow coordination, and economic feasibility within competitive project environments.

In the preconstruction phases, the methodology shifts BIM coordination from purely geometric clash detection to semantic planning. Coordinators and safety managers must collaborate to identify BIM families requiring high-fidelity tracking and map specific safety risks (e.g., HAV) to these objects. This integrates safety planning directly into the digital model rather than keeping it in siloed documents. The construction phase implementation requires coordination between field superintendents, safety supervisors, and technology coordinators to manage the daily operations of both robotic scanning activities and worker-worn sensor monitoring. Superintendents need clear protocols for scheduling robot deployment activities that complement rather than interfere with active construction operations.

BIM2RDT transforms robot missions from simple documentation exercises into intelligent, context-aware inspection protocols. Traditional robotic scanning follows pre-programmed paths that treat the site as unknown territory, requiring exhaustive coverage to capture all conditions. In contrast, BIM2RDT-equipped UGVs leverage semantic expectations from the BIM to focus scanning resources on high-value targets—objects likely to have deviated from design, temporary elements requiring verification, or zones with active safety-critical work. In other words, since perception is BIM-conditioned (target families and ROIs) and alignment is stabilized by object-adaptive priors, robot missions can shift from broad, exhaustive scanning towards targeted and repeatable verification passes.

For project managers and safety personnel, the primary decision impact is the ability to move from reactive interpretation of raw photos/point clouds towards actionable, location-grounded signals. Current practice separates safety monitoring (manual observations, incident

reports) from progress tracking (periodic surveying, photo documentation). BIM2RDT unifies these streams within the digital twin, enabling managers to make risk-informed scheduling decisions. For example, deviations can trigger targeted re-inspections or coordination checks, while safety events (e.g., threshold exceedances) can trigger intervention tasks and influence where robots are routed to avoid interfering with active high-risk work zones.

6. Conclusion

This paper introduced BIM2RDT, a context-aware, AI-powered framework that transforms static Building Information Models into dynamic, robot-ready digital twins with an emphasis on safety. By integrating BIM data, robot-collected point clouds, and IoT sensor streams for activity monitoring, the framework provides a robust solution for real-time site understanding. The proposed Semantic-Gravity ICP (SG-ICP) algorithm, which leverages LLM-based reasoning to inform physical priors, was shown to significantly improve point cloud registration accuracy, achieving RMSE reductions of up to 97% over standard methods and ensuring physically plausible alignments. Furthermore, the integration of HAV monitoring demonstrated the framework's capability to map safety-critical events to the digital twin, supporting safety monitoring and risk-aware planning.

Despite its contributions, the work has limitations. First, the experiments were conducted in controlled settings, and the framework's scalability to large, highly dynamic construction sites with diverse and unanticipated objects remains to be validated. This includes practical sources of variability that were not captured in the proof-of-concept evaluation (e.g., severe occlusions, dust/lighting changes, frequent reconfiguration of temporary works, and intermittent connectivity that can affect data transfer and update frequency). Furthermore, the stress testing of the open-vocabulary detection needs to be further verified. Given that BIMs have unique nomenclature between jobsites, an accurate mapping between them all is essential to the framework's success. The HAV monitoring demonstration uses pre-recorded vibration signals replayed in simulation rather than data collected during actual tool operation on site. Also, the framework's dependence on high-quality BIM models as a priori maps may limit applicability in projects with incomplete or poorly structured digital models.

Second, our current "agentic" scope is intentionally bounded for two bounded reasoning tasks: assigning BIM-family upright priors for SG-ICP and mapping detected labels to IFC GUIDs for synchronization. The system employs LLMs for semantic reasoning and orchestrates multi-step perception-to-update workflows, but it does not exhibit autonomous long-horizon planning, goal decomposition, or adaptive replanning in response to environmental feedback—capabilities typically associated with fully agentic systems. Thus, with proper referencing for adopting the perspective that Agentic AI is characterized by iterative reasoning and tool use rather than zero-shot execution, "agentic" here describes the system's use of LLM-based semantic reasoning to condition perception and updates, rather than claiming full agent autonomy.

Third, the safety component is demonstrated using HAV as an ISO-standardized, measurable signal, but it is not intended to represent comprehensive site safety coverage; broader hazard categories (e.g., proximity risks, fall hazards, dynamic exclusion zones) and longitudinal validation of intervention effectiveness remain future work. Extending this to multi-worker, multi-hazard scenarios (e.g., simultaneous HAV, heat stress, fall risk, noise exposure across dozens of workers) introduces sensor fusion challenges, communication bandwidth requirements, and computational complexity not addressed in the current work.

Future work will focus on conducting extensive field trials on active construction sites to validate robustness, developing error-correction mechanisms for imperfect BIM inputs, and optimizing algorithms for edge computing to improve real-time performance and scalability. Research should also focus on expanding the AI orchestrator to handle broader sensor fusion scenarios (extending to an agentic layer that already assigns semantic priors for SG-ICP). For example, for the HAV detection scenario presented here as a proof-of-concept, after vibration metric computation (e.g., frequency-weighted RMS acceleration a_{hv} compared to ISO thresholds), an agent can process IoT data from wearables to classify activities, inferring labels like "prolonged HAV exposure" based on signal patterns and contextual cues from robot observations. A semantic mapping agent can then cross-reference these with BIM elements and using LLM-based reasoning, aligns detected activities to *IfcTask* instances.

CRediT authorship contribution statement

Reza Akhavian: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mani Amani:** Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Johannes Mootz:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Behrad Beheshti:** Writing – original draft, Validation, Software, Methodology, Data curation. **Robert Ashe:** Writing – original draft, Visualization, Validation, Software, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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